

UPPERBOUND

1. $f(n) = 6n + 3$

Solution

To find upper bound of $f(n)$, we have to find c and n_0 such that $0 \leq f(n) \leq c \times g(n)$ for all $n \geq n_0$

$$0 \leq f(n) \leq c \times g(n)$$

$$0 \leq 6n + 3 \leq c \times g(n)$$

$$0 \leq 6n + 3 \leq 6n + 3n, \text{ for all } n \geq 1 \text{ (There can be such infinite possibilities)}$$

$$0 \leq 6n + 3 \leq 9n$$

$$\text{So, } c = 9 \text{ and } g(n) = n, n_0 = 1$$

Tabular Approach

$$0 \leq 6n + 3 \leq c \times g(n)$$

$$0 \leq 6n + 3 \leq 7n$$

Now, manually find out the proper n_0 , such that $f(n) \leq c.g(n)$

n	$f(n) = 6n + 3$	$c.g(n) = 7n$
1	9	7
2	15	14
3	21	21
4	27	28
5	33	35

From Table, for $n \geq 3$, $f(n) \leq c \times g(n)$ holds true. So, $c = 7$, $g(n) = n$ and $n_0 = 3$, There can be such multiple pair of (c, n_0)

$$f(n) = O(g(n)) = O(n) \text{ for } c = 9, n_0 = 1$$

$$f(n) = O(g(n)) = O(n) \text{ for } c = 7, n_0 = 3$$

2. $f(n) = 3n^2 + 2n + 4$

Solution:

To find upper bound of $f(n)$, we have to find c and n_0 such that $0 \leq f(n) \leq c \times g(n)$ for all $n \geq n_0$

$$0 \leq f(n) \leq c \times g(n)$$

$$0 \leq 3n^2 + 2n + 4 \leq c \times g(n)$$

$$0 \leq 3n^2 + 2n + 4 \leq 3n^2 + 2n^2 + 4n^2,$$

for all $n \geq 1$:

$$0 \leq 3n^2 + 2n + 4 \leq 9n^2$$

$$0 \leq 3n^2 + 2n + 4 \leq 9n^2$$

So, $c = 9$, $g(n) = n^2$ and $n_0 = 1$

Tabular Approach

$$0 \leq 3n^2 + 2n + 4 \leq c \cdot g(n)$$

$$0 \leq 3n^2 + 2n + 4 \leq 4n^2$$

Now, manually find out the proper n_0 , such that $f(n) \leq c \cdot g(n)$

n	$f(n) = 3n^2 + 2n + 4$	$c \cdot g(n) = 4n^2$
1	9	4
2	20	16
3	37	36
4	60	64
5	89	100

From Table, for $n \geq 4$, $f(n) \leq c \times g(n)$ holds true. So, $c = 4$, $g(n) = n^2$ and $n_0 = 4$. There can be such multiple pair of (c, n_0)

$$f(n) = O(g(n)) = O(n^2) \text{ for } c = 9, n_0 = 1$$

$$f(n) = O(g(n)) = O(n^2) \text{ for } c = 4, n_0 = 4$$

3. $f(n) = 2n^3 + 4n + 5$

Solution:

To find upper bound of $f(n)$, we have to find c and n_0 such that $0 \leq f(n) \leq c \times g(n)$ for all $n \geq n_0$

$$0 \leq f(n) \leq c \cdot g(n)$$

$$0 \leq 2n^3 + 4n + 5 \leq c \times g(n)$$

$$0 \leq 2n^3 + 4n + 5 \leq 2n^3 + 4n^3 + 5n^3, \text{ for all } n \geq 1$$

$$0 \leq 2n^3 + 4n + 5 \leq 11n^3$$

$$\text{So, } c = 11, g(n) = n^3 \text{ and } n_0 = 1$$

Tabular Approach

$$0 \leq 2n^3 + 4n + 5 \leq c \times g(n)$$

$$0 \leq 2n^3 + 4n + 5 \leq 3n^3$$

Now, manually find out the proper n_0 , such that $f(n) \leq c \times g(n)$

n	$f(n) = 2n^3 + 4n + 5$	$c \cdot g(n) = 3n^3$
1	11	3
2	29	24
3	71	81
4	149	192

From Table, for $n \geq 3$, $f(n) \leq c \times g(n)$ holds true. So, $c = 3$, $g(n) = n^3$ and $n_0 = 3$. There can be such multiple pair of (c, n_0)

$$f(n) = O(g(n)) = O(n^3) \text{ for } c = 11, n_0 = 1$$

$$f(n) = O(g(n)) = O(n^3) \text{ for } c = 3, n_0 = 3 \text{ and so on.}$$

LOWERBOUND

1. $f(n) = 6n + 3$

Solution:

To find lower bound of $f(n)$, we have to find c and n_0 such that $0 \leq c \cdot g(n) \leq f(n)$ for all $n \geq n_0$

$$0 \leq c \cdot g(n) \leq f(n)$$

$$0 \leq c \cdot g(n) \leq 6n + 3$$

$$0 \leq 6n \leq 6n + 3 \rightarrow \text{true, for all } n \geq n_0$$

$$0 \leq 5n \leq 6n + 3 \rightarrow \text{true, for all } n \geq n_0$$

Above both inequalities are true and there exists such infinite inequalities. So,

$$f(n) = \Omega(g(n)) = \Omega(n) \text{ for } c = 6, n_0 = 1$$

$$f(n) = \Omega(g(n)) = \Omega(n) \text{ for } c = 5, n_0 = 1$$

2. $f(n) = 2n^3 + 4n + 5$

Solution:

To find lower bound of $f(n)$, we have to find c and n_0 such that $0 \leq c \cdot g(n) \leq f(n)$ for all $n \geq n_0$

$$0 \leq c \cdot g(n) \leq f(n)$$

$$0 \leq c \cdot g(n) \leq 2n^3 + 4n + 5$$

$$0 \leq 2n^3 \leq 2n^3 + 4n + 5 \rightarrow \text{true, for all } n \geq 1$$

$$0 \leq n^3 \leq 2n^3 + 4n + 5 \rightarrow \text{true, for all } n \geq 1$$

Above both inequalities are true and there exists such infinite inequalities.

$$\text{So, } f(n) = \Omega(g(n)) = \Omega(n^3) \text{ for } c = 2, n_0 = 1$$

$$f(n) = \Omega(g(n)) = \Omega(n^3) \text{ for } c = 1, n_0 = 1$$

TIGHTBOUND

1. $f(n) = 6n + 3$

Solution:

To find tight bound of $f(n)$, we have to find c_1 , c_2 and n_0 such that, $0 \leq c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$ for all $n \geq n_0$

$$0 \leq c_1 \cdot g(n) \leq 6n + 3 \leq c_2 \cdot g(n)$$

$$0 \leq 5n \leq 6n + 3 \leq 9n, \text{ for all } n \geq 1$$

Above inequality is true and there exists such infinite inequalities.

$$\text{So, } f(n) = \Theta(g(n)) = \Theta(n) \text{ for } c_1 = 5, c_2 = 9, n_0 = 1$$

2. $f(n) = 3n^2 + 2n + 4$

Solution:

To find tight bound of $f(n)$, we have to find c_1 , c_2 and n_0 such that, $0 \leq c_1 \times g(n) \leq f(n) \leq c_2 \times g(n)$ for all $n \geq n_0$

$$0 \leq c_1 \times g(n) \leq 3n^2 + 2n + 4 \leq c_2 \times g(n)$$

$$0 \leq 3n^2 \leq 3n^2 + 2n + 4 \leq 9n^2, \text{ for all } n \geq 1$$

Above inequality is true and there exists such infinite inequalities. So,

$$f(n) = \Theta(g(n)) = \Theta(n^2) \text{ for } c_1 = 3, c_2 = 9, n_0 = 1$$

3. $f(n) = 2n^3 + 4n + 5$

Solution:

To find tight bound of $f(n)$, we have to find c_1 , c_2 and n_0 such that, $0 \leq c_1 \times g(n) \leq f(n) \leq c_2 \times g(n)$ for all $n \geq n_0$

$$0 \leq c_1 \times g(n) \leq 2n^3 + 4n + 5 \leq c_2 \times g(n)$$

$$0 \leq 2n^3 \leq 2n^3 + 4n + 5 \leq 11n^3, \text{ for all } n \geq 1$$

Above inequality is true and there exists such infinite inequalities. So,

$$f(n) = \Theta(g(n)) = \Theta(n^3) \text{ for } c_1 = 2, c_2 = 11, n_0 = 1$$